

AMERICAN UNIVERSITY
OF MADABA
الجامعـــــة الأميركيّـــــة
في مادبـا

Lane Detection System

Name: Ghazal Helal, Ayat Omar, Zaid Eltabatebi

ID: 2020001, 2020091, 2110012

Instructor: Dr. Ahmad Daraiseh

Course: Fundamentals of Artificial Intelligence

Lane Detection System

Abstract:

The project is a computer vision-based system for processing pictures and video frames taken by a vehicle-mounted camera. The objective is processing images and videos to identify and mark lane lines and calculate the distance for the car in the front. The main goal is to locate and monitor lane markings on the road, approaching the model-based and learning-based methods which is essential for the advancement of advanced driver assistance and autonomous driving technologies

Introduction:

Accurately detecting and tracking lane lines is essential for autonomous driving and computer vision applications. The goal of this project is to develop a reliable pipeline for lane identification that receives input from a camera, processes it, adjusts for distortions, changes the image's perspective, employs thresholding techniques, and, it employs various libraries such as NumPy, OpenCV, Matplotlib, Docopt, IPython, and MoviePy in the end to display the identified lane lines. The final product offers an accurate depiction of the geometry of the road, providing important data for driver assistance or autonomous navigation.

Important Elements:

- **Calibration of Cameras:**

Using a collection of calibration photos, the system first calibrates the camera. Lens distortions are adjusted for in this phase.

- **Perspective Shift:**

By transforming the image into a top-down view, this transformation streamlines the lane recognition process and facilitates lane line identification and tracking.

- **Limiting:**

Different thresholding strategies are used to improve the visibility of the lane lines in the pictures. this stage entails highlighting important characteristics and hiding unneeded details.

- **Identifying Lane Lines:**

The algorithm used to find and fit curves to the lanes, this combines color space analysis, gradient-based approaches, and sliding window techniques.

- **Illustration:**

Visualizations of the identified lane lines superimposed on the original photos or video frames make up the final output.

- **Assessment of Performance:**

Performance indicators, such as the precision of lane detection, adaptability to changing lighting, and fluidity of lane tracking in video sequences, are used to assess the system's effectiveness.

Use Cases and Applications:

This project's lane detection pipeline has immediate implications in autonomous cars, where a thorough grasp of the surrounding road environment is necessary for dependable and safe navigation. It can also be utilized in sophisticated DAS systems to improve the consciousness of the driver and help them maintain appropriate lane discipline.

The project lays the foundation for safer and more effective transportation by utilizing computer vision techniques to help construct intelligent systems that can comprehend and navigate real-world road settings.

Use cases	Description	Advantages
Autonomous Driving	The primary use case involves the integration of the lane detection system into autonomous vehicles. The system aids in identifying and tracking lane lines, allowing the vehicle to navigate within lanes autonomously.	Enhances the vehicle's ability to stay within designated lanes, contributing to safer and more reliable autonomous driving.
Driver Assistance Systems	The lane detection system can be integrated into driver assistance technologies to provide real-time feedback to human drivers. This includes warnings or alerts when the vehicle deviates from the lane.	Improves driver awareness and reduces the risk of unintentional lane departures, promoting overall road safety.
Traffic Monitoring and Management	The project can be employed for real-time traffic monitoring on roads and highways. The system detects and analyzes lane usage patterns, aiding in traffic flow management.	Enables authorities to optimize traffic signal timings, implement dynamic lane assignments, and enhance overall traffic management.

Smart Parking Systems	Integrating the lane detection system with parking management can enhance smart parking solutions. The system can assist drivers in aligning their vehicles within parking spaces.	Improves parking accuracy, reduces the risk of collisions, and enhances the efficiency of parking spaces in urban environments.
Augmented Reality Navigation	Lane detection results can be integrated into augmented reality (AR) navigation systems. AR displays can overlay lane information directly onto the driver's view.	Provides intuitive and context-aware navigation information, enhancing the driver's situational awareness.

Peas for our system:

1) Agent: The FindLaneLines class is the main agent in this system. It makes judgments on how to process image or video frames in order to detect and visualize lane lines. It encompasses the whole pipeline for lane detection.

2) Environment: The stream of pictures or video frames from a car-mounted camera is the environment in this instance. In order to identify and follow lane lines on the road, the agent analyzes each frame.

3) Actuators: The procedures and methods in the FindLaneLines class serve as the actuators in this system. These consist of lane line detection, thresholding, perspective adjustments, image undistorting, and visualizing the outcomes.

4) Sensors: The camera, which records frames for pictures or videos, is the main sensor. Video input is handled via the VideoFileClip class from moviepy, while image files are read using the mpimg.imread method. By correcting for distortions, the camera calibration procedure also functions as a kind of sensing

Formulating for problem:

1) Initial state: A raw image or frame that the camera captured is the starting condition. After that, the raw data is processed via a pipeline, with the first stage being camera calibration to remove distortion and get the image ready for further processing.

2) Actions: The steps the agent takes to process each frame are called actions. These consist of perspective transformation, thresholding, lane line recognition, camera calibration, and visualization. The agent's flow of actions is represented using the forward method.

3) Transition Model: explains the process by which the agent's activities cause the system to change states. The set of image processing operations that are applied to every frame in this project is known as the transition model.

4) Performance Measurement: The accuracy and recall of lane line recognition, the fluidity of video lane tracking, and the pipeline's computing efficiency are examples of common performance measures.

5) Goal state: The processed image or video frame with successfully recognized and visible lane lines is the goal state. With the goal of offering reliable and consistent lane information, the agent can be used for applications like as driver assistance or autonomous driving.

Area of interest:

The creation and improvement of a computer vision-based lane detection system for automobiles is the main focus of the project. This entails improving thresholding, lane identification, perspective transformation, and camera calibration algorithms. Real-time processing, environmental robustness, integration with autonomous systems, and extensive testing and validation are important factors to take into account.

Conclusion:

The crucial challenge of lane recognition and visualization in a vehicle-mounted camera system is satisfactorily addressed by the project. The system accomplishes precise and reliable lane line identification in both picture and video frames by putting in place an extensive pipeline that consists of camera calibration, perspective transformation, thresholding, and lane recognition. Code structure and maintainability are improved by the usage of custom classes, external library integration, and modular design.

Key achievements include the optimization of camera calibration and lane detection techniques for adaptability to diverse driving conditions. Real-time processing considerations, environmental robustness, and integration potential with autonomous systems underscore the project's relevance to practical applications in autonomous driving and advanced driver assistance.